



SUMMARY

Experienced Data Scientist and Machine Learning Engineer with **4+ years** of hands-on expertise in building and deploying AI solutions across finance and telecom industries. Skilled in developing predictive models, NLP systems, anomaly detection, time-series forecasting, and deep learning using Python, TensorFlow, PyTorch, and SQL. Proven ability to design real-time ETL pipelines, automate fraud detection, and enhance trading strategies for improved compliance and risk management. Proficient in creating interactive dashboards using Power BI and Tableau to support data-driven decisions. Strong team player with a track record of integrating AI into core business operations to boost efficiency, accuracy, and customer experience.

TECHNICAL SKILLS

Programming & web :	Python, SQL, R, JavaScript, PySpark, Bash, Shell,
ML & Analytics:	TensorFlow, PyTorch, Scikit-learn, XGBoost, LightGBM, CatBoost, AutoML, Reinforcement Learning, GANs, Neural Networks (CNN, RNN, LSTM), BERT, LoRA, QLoRA, Prophet, ARIMA, SHAP, LIME, Captum, A/B Testing
Generative AI & LLMs:	OpenAI SDK, LangChain, LlamaIndex, Transformers (Hugging Face), Guardrails, LangGraph, AutoGen, CrewAI, LangSmith, LangFuse, Prompt Engineering, Agentic Workflows, Retrieval- Augmented Generation (RAG) Models, Vector Search (FAISS, Pinecone, Weaviate, ChromaDB), Computer Vision, DALL-E, GPT-4, Claude, Gemini, BERT
Data Engineering & ETL:	Regression & Classification, Clustering (K-Means, DBSCAN), PCA, Time Series Forecasting, Feature Engineering, Statistical Analysis, Data Visualization (Matplotlib, Seaborn, Plotly, Power BI, Tableau, Excel) Data Wrangling/Cleaning, CNN, RNN, Deep Learning, A/B Testing
Cloud& Deployment:	ML, Azure Cognitive Services, AWS SageMaker, AWS Lambda, S3, EKS, Redshift, Bedrock, GCP Vertex AI, Docker, Kubernetes, Helm, Flask, FastAPI, MLflow, CI/CD, Jenkins, Terraform, Grafana, Co-Pilot Studio
Databases & Storage:	MySQL, PostgreSQL, MongoDB (NoSQL), Snowflake, Delta Lake, Azure Data Lake (ADLS Gen2), BigQuery, Vector Databases (Pinecone, ChromaDB)
Data Visualization:	Power BI, Tableau, Matplotlib, Seaborn, Plotly, Streamlit, Grafana, Geospatial & Interactive Dashboards, Data Storytelling

PROFESSIONAL EXPERIENCE

Data Scientist with AI/ML
Charles Schwab, TX

May 2024 – Present

- Architected a Portfolio Risk Monitoring and Insights Platform on Azure Databricks, Synapse, and Azure ML, leveraging Transformer-based NLP for real-time sentiment extraction and market signal generation, boosting investment strategy accuracy by 35%.
- Designed multi-agent AI workflows using LangGraph and CrewAI for automated document parsing, market summarization, and client alerting, reducing manual review cycles by 60%.
- Built a Monte Carlo and Reinforcement Learning hybrid model for portfolio risk calibration and adaptive hedging, improving capital efficiency by 25%.
- Deployed a graph-driven fraud intelligence system combining GNNs, Isolation Forest, and Autoencoders to detect relational anomalies in under 50 ms across streaming data processed via Kafka and PySpark.
- Implemented retrieval-augmented compliance review pipelines (RAG) using Azure Cognitive Search, LangChain Guardrails, and GPT-4 to extract and validate regulatory fields with high factual integrity.
- Orchestrated continuous training, validation, and deployment through MLflow, Kubernetes, and Azure ML Pipelines, achieving reproducible releases and measurable MLOps reliability gains.
- Delivered Power BI and Grafana dashboards visualizing portfolio exposure, sentiment drift, and SHAP-based model explainability, improving risk visibility for leadership teams.
- Integrated drift-aware model governance using Azure Monitor, Data Drift Detector, and custom lineage tracking, enabling automated retraining triggers and improving model stability by 40%.

- Built a document intelligence solution using AWS Textract, Comprehend, and BERT-based NER models to automate KYC extraction and onboarding, cutting manual validation time by 65%.
- Designed credit risk scoring models leveraging XGBoost, LightGBM, and Scikit-learn pipelines deployed on AWS SageMaker, improving loan approval accuracy and reducing default rates by 18%.
- Implemented NLP-based regulatory summarization using Hugging Face Transformers and spaCy, automating AML and compliance document review with a 40% reduction in turnaround time.
- Created behavioral segmentation pipelines using K-Means, HDBSCAN, and LSTM-based time series models, enabling personalized offers and dynamic customer targeting.
- Automated data ingestion and transformation workflows using AWS Glue, Lambda, Step Functions, and Redshift Spectrum, ensuring consistent schema enforcement and improving data availability by 90%.
- Developed a rule-based AML screening engine integrating entity resolution algorithms and similarity scoring techniques to detect high-risk relationships and anomalous transactions.
- Established MLOps practices using SageMaker Model Monitor, CloudWatch, and Git-based CI/CD, enabling version tracking and proactive drift detection across deployed models.
- Collaborated with risk and compliance teams to implement an explainability layer using SHAP and LIME visualizations in SageMaker Studio, enhancing transparency and regulator trust in AI-driven decision models.

- Worked on a Data Pipeline Development & Reporting project for a Telecom client, contributing to the design of end-to-end ETL pipelines using Python, SQL, and Apache Spark to ingest and transform data, reducing manual data preparation time by 25%.
- Supported the creation of data warehouse schemas and models in Snowflake and PostgreSQL, applying normalization and indexing techniques that improved query performance by 30%.
- Implemented data quality checks (duplicate removal, null handling, schema validation) which increased dataset accuracy and consistency for reporting by 20%.
- Automated workflow scheduling using Apache Airflow and generated Excel-based reports that improved KPI visibility for customer activity and billing accuracy, cutting report turnaround time by 40%.
- Collaborated with senior engineers to integrate data ingestion pipelines from CSV, JSON, and API sources, leveraging Git for version control and following Agile sprint practices, gaining early exposure to enterprise-grade data engineering workflows.

EDUCATION

CERTIFICATIONS

AWS CERTIFIED DATA ENGINEER-ASSOCIATE
MICROSOFT CERTIFIED- AZURE DATA SCIENTIST ASSOCIATE AWS
CERTIFIED MACHINE LEARNING -SPECIALITY MICROSOFT
CERTIFIED-AZURE AI ENGINEER ASSOCIATE

PROJECTS

Loan Default Prediction

- Built an end-to-end loan default prediction system using Python and machine learning to identify high-risk borrowers with over 90% accuracy. The project included data cleaning, feature engineering, and training multiple classification models including Logistic Regression, Random Forest, and XGBoost.
- A real-time prediction interface was developed using Streamlit, allowing users to input loan application details and receive instant risk feedback. Additionally, Power BI dashboards were created to visualize key patterns in defaults across loan grade, term, interest rates, and employment length.
- Enabled deeper insights into borrower behavior, supporting informed loan approval decisions.

Customer Churn Prediction using Machine Learning

- Built a predictive model to identify telecom customers likely to churn using billing, usage, and demographic data. Applied preprocessing, PCA, and VIF to prepare features, and trained models like Random Forest and XGBoost.
- Optimized performance using Grid SearchCV and evaluated with ROC-AUC and confusion matrix. Delivered insights to support retention strategies through visual reports.